INCH-POUND

AN846 Rev 10
12 March 2020
SUPERSEDING
AN846 Rev 9
3 October 2013

DETAIL SPECIFICATION SHEET

ELBOW, HOSE, 45 DEGREE

Inactive for new design after 7 June 2011. For new design, use SAE-AS5187.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843.

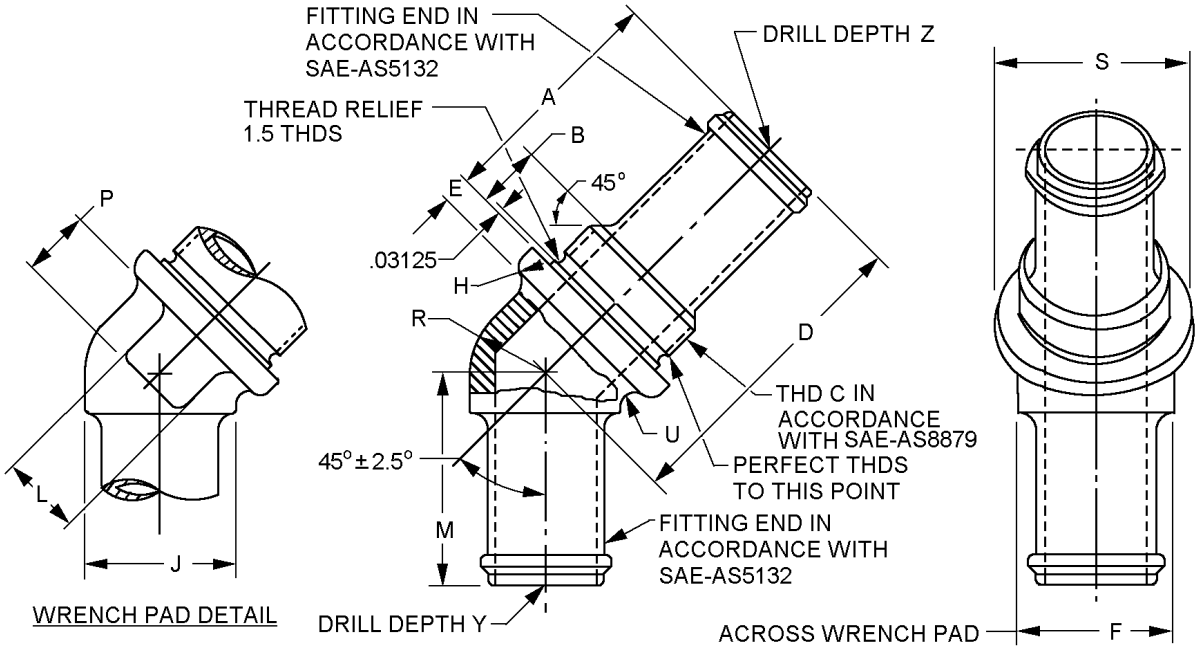


FIGURE 1. 45° elbow dimensions and configuration.

AMSC N/A

FSC 4730

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Dash number	Hose ID inches (mm)	A inches (mm)	B inches (mm)	Thread C Ref SAE-AS8879
4	.250 (6.35)	1.938 (49.23)	.438 (11.13)	.4375-20UNJF-3A
6	.375 (9.53)	1.969 (50.01)	.469 (11.91)	.5625-18UNJF-3A
8	.500 (12.70)	2.000 (50.80)	.500 (12.70)	.7500-16UNJF-3A
10	.625 (15.88)	2.031 (51.59)	.531 (13.49)	.8750-14UNJ-3A
12	.750 (19.05)	2.063 (52.40)	.563 (14.30)	1.0625 -12UNJ-3A
16	1.000 (25.40)	2.063 (52.40)	.563 (14.30)	1.3125-12UNJ-3A
20	1.250 (31.75)	2.063 (52.40)	.563 (14.30)	1.6250-12UNJ-3A
24	1.500 (38.10)	2.063 (52.40)	.563 (14.30)	1.8750-12UNJ-3A

Dash number	D .047 (1.19) -.000 inches (mm)	E inches (mm)	F inches (mm)	H Radius inches (mm)	J Diameter inches (mm)
4	2.266 (57.56)	.156 (3.96)	.438 (11.13)	.063 (1.60)	.438 (11.13)
6	2.406 (61.11)	.188 (4.78)	.563 (14.30)	.063 (1.60)	.563 (14.30)
8	2.500 (63.50)	.219 (5.56)	.750 (19.05)	.094 (2.39)	.750 (19.05)
10	2.641 (67.08)	.250 (6.35)	.875 (22.23)	.094 (2.39)	.875 (22.23)
12	2.703 (68.66)	.250 (6.35)	1.063 (27.00)	.094 (2.39)	1.063 (27.00)
16	2.750 (69.85)	.250 (6.35)	1.313 (33.35)	.125 (3.18)	1.313 (33.35)
20	2.938 (74.63)	.281 (7.14)	1.625 (41.28)	.125 (3.18)	1.625 (41.28)
24	3.000 (76.20)	.281 (7.14)	1.875 (47.63)	.125 (3.18)	1.875 (47.63)

Dash number	L Approx. inches (mm)	M +.047 (1.19) -.000 inches (mm)	P Approx. inches (mm)	R Radius inches (mm)
4	.250 (6.35)	1.609 (40.87)	.219 (5.56)	.219 (5.56)
6	.375 (9.53)	1.656 (42.06)	.281 (7.14)	.281 (7.14)
8	.438 (11.13)	1.688 (42.88)	.344 (8.75)	.375 (9.53)
10	.500 (12.70)	1.734 (44.04)	.438 (11.13)	.438 (11.13)
12	.563 (14.30)	1.766 (44.86)	.500 (12.70)	.531 (13.49)
16	.688 (17.48)	1.828 (46.43)	.563 (14.30)	.656 (16.66)
20	1.000 (25.40)	1.922 (48.82)	.688 (17.48)	.813 (20.65)
24	1.125 (28.58)	1.969 (50.01)	.750 (19.05)	.938 (22.83)

See notes at end of figure.

FIGURE 1. 45° elbow dimensions and configuration - Continued.

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Dash number	S Diameter inches (mm)	U Radius inches (mm)	Y +.047 (1.19) -.000 inches (mm)	Z +.047 (1.19) -.000 inches (mm)
4	.625 (15.88)	.063 (1.60)	1.641 (41.68)	2.297 (58.34)
6	.813 (20.65)	.094 (2.39)	1.719 (43.66)	2.469 (62.71)
8	1.000 (25.40)	.094 (2.39)	1.766 (44.86)	2.578 (65.48)
10	1.250 (31.75)	.125 (3.18)	1.848 (46.94)	2.750 (69.85)
12	1.438 (36.53)	.125 (3.18)	1.891 (48.03)	2.828 (71.83)
16	1.688 (42.88)	.125 (3.18)	2.016 (51.21)	2.938 (74.63)
20	2.063 (52.40)	.188 (4.48)	2.141 (54.38)	3.156 (80.16)
24	2.375 (60.33)	.188 (4.48)	2.250 (57.15)	3.281 (83.34)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges .and remove all hanging burrs and slivers.
5. Machined surfaces shall be finished to 125 µ-in in Ra, forged surfaces shall be 250 µ-in Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. 45° elbow dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Materials and finishes shall be in accordance with SAE-AS4843, see table I for material and finish code.

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TABLE I. Material and finish codes.

Material and finish code	Material	Protective chemical finish 3/ 4/
No code 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	No finish.
BC 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3.
D 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	Anodize in accordance with SAE-AMS2472 or MIL-A-8625, type II, class 2, dye blue, duplex seal in accordance with procurement specification.
DV 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
J	Type 304 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5639.	Passivate in accordance with SAE-AMS2700, type VI or VII.
K	Type 316 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5648.	Passivate in accordance with SAE-AMS2700, type VI or VII.

See notes at end of table

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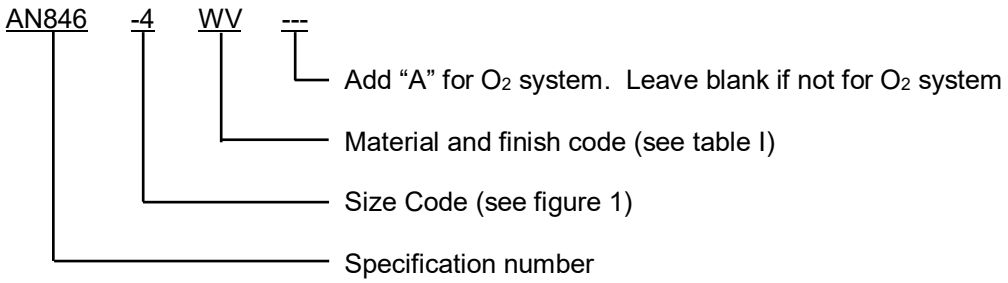
TABLE I. Material and finish codes - Continued.

Material and finish code	Material	Protective chemical finish 3/ 4/
R	Type 321 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5645.	Passivate in accordance with SAE-AMS2700, type VI or VII.
W 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	Anodize in accordance with SAE-AMS2472 or MIL-A-8625, type II, class 2, dye brown similar to color SAE-AMS-STD-595/10080, duplex seal in accordance with procurement specification.
WV 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
T 3/	Titanium	Anodize in accordance with SAE-AMS2488, type 2.

1/ Material code was dash on previous revisions, changed to agree with SAE-ARP1590.
2/ Aluminum code D is inactivated, use code W.
3/ Titanium and cadmium shall not be used in oxygen systems.
4/ Cadmium shall not be used in potable water systems.

Oxygen systems for aerospace, shipboard, and ground support equipment. Parts for use on oxygen systems shall be identified in the PIN as code “A” and shall be furnished cleaned, packaged, and labeled in accordance with SAE-AS611 to a process approved by the user.

Part or Identifying Number (PIN): The PIN consists of the letter “AN” the specification sheet number, dash number for ID hose sizes, and a code letter for material type and finish, and blank or “A” for oxygen cleaning requirements. Unassigned PIN’s shall not be used.



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PIN examples:

AN846-8WP indicates a 45° elbow, hose to hose, .500 inch (12.70 mm) ID, aluminum alloy 7075-T73.

AN846-8WV indicates a 45° elbow, hose to hose, .500 inch (12.70 mm) ID, aluminum alloy 7075-T73 finish with high purity aluminum.

AN844-8WA indicates a 45° elbow, hose to hose, .500 inch (12.70 mm) ID, aluminum alloy 7075-T73 for use on oxygen systems.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Table II provides a detailed cross-reference of AN846 PINs and replacement SAE-AS5187 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE II. Cross-reference data. 1/

AN PIN Inactive part	Hose Size	Tube Size	AS PIN 2/ Active part
AN846-4	0.250	0.250	AS5187B04
AN846-4D	0.250	0.250	AS5187W04
AN846-4J	0.250	0.250	AS5187J04
AN846-4K	0.250	0.250	AS5187K04
AN844-4R	0.250	0.250	AS5187R04
AN844-4S	0.250	0.250	AS5187R04
AN846-4T	0.250	0.250	None
AN846-4W	0.250	0.250	AS5187W04
AN846-6	0.375	0.375	AS5187B06
AN846-6D	0.375	0.375	AS5187W06
AN846-4J	0.375	0.375	AS5187J06
AN846-4K	0.375	0.375	AS5187K06
AN846-6R	0.375	0.375	AS5187R06
AN846-6S	0.375	0.375	AS5187R06
AN846-4T	0.375	0.375	None
AN846-6W	0.375	0.375	AS5187W06
AN846-8	0.500	0.500	AS5187B08
AN846-8D	0.500	0.500	AS5187W08
AN846-8J	0.500	0.500	AS5187J08
AN846-8K	0.500	0.500	AS5187K08
AN846-8R	0.500	0.500	AS5187R08
AN846-8S	0.500	0.500	AS5187R08
AN846-8T	0.500	0.500	None
AN846-8W	0.500	0.500	AS5187W08
AN846-10	0.625	0.625	AS5187B10
AN846-10D	0.625	0.625	AS5187W10
AN846-10J	0.625	0.625	AS5187J10
AN846-10K	0.625	0.625	AS5187K10
AN846-10R	0.625	0.625	AS5187R10
AN846-10S	0.625	0.625	AS5187R10
AN846-10T	0.625	0.625	None
AN846-10W	0.625	0.625	AS5187W10

See notes at end of table.

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TABLE II. Cross-reference data - Continued. 1/

AN PIN Inactive part	Hose Size	Tube Size	AS PIN 2/ Active part
AN846-12	0.750	0.750	AS5187B12
AN846-12D	0.750	0.750	AS5187W12
AN846-12J	0.750	0.750	AS5187J12
AN846-12K	0.750	0.750	AS5187K12
AN846-12R	0.750	0.750	AS5187R12
AN846-12S	0.750	0.750	AS5187R12
AN846-12T	0.750	0.750	None
AN846-12W	0.750	0.750	AS5187W12
AN846-16	1.000	1.000	AN5187B16
AN846-16D	1.000	1.000	AS5187W16
AN846-16J	1.000	1.000	AS5187J16
AN846-16K	1.000	1.000	AS5187K16
AN846-16R	1.000	1.000	AS5187R16
AN846-16S	1.000	1.000	AS5187R16
AN846-16T	1.000	1.000	None
AN846-16W	1.000	1.000	AS5187W16
AN846-20	1.250	1.250	AS5187B20
AN846-20D	1.250	1.250	AS5187W20
AN846-20J	1.250	1.250	AS5187J20
AN846-20K	1.250	1.250	AS5187K20
AN846-20R	1.250	1.250	AS5187R20
AN846-20S	1.250	1.250	AS5187R20
AN846-20T	1.250	1.250	None
AN846-20W	1.250	1.250	AS5187W20
AN846-24	1.500	1.500	AS5187B24
AN846-24D	1.500	1.500	AS5187W24
AN846-24J	1.500	1.500	AS5187J24
AN846-24K	1.500	1.500	AS5187K24
AN846-24R	1.500	1.500	AS5187R24
AN846-24S	1.500	1.500	AS5187R24
AN846-24T	1.500	1.500	None
AN846-24W	1.500	1.500	AS5187W24

1/ For new design use material designator R and W.

2/ SAE “B” designator for copper alloy parts can be either “no finish” or “cadmium plated”. This makes determination of hazardous materials more difficult and disposing of hazardous materials more laborious.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to SAE-AS4843, this document references the following:

MIL-A-8625	SAE-AMS-QQ-S-763	SAE-AMS4614
MIL-DTL-83488	SAE-AMS-STD-595/10080	SAE-AMS5639
ASME B46.1	SAE-AMS2472	SAE-AMS5645
ASTM B124/B124M	SAE-AMS2488	SAE-AMS5648
ASTM B138/B138M	SAE-AMS2700	SAE-ARP1590
SAE-AMS-QQ-A-225/6	SAE-AMS4124	SAE-AS611
SAE-AMS-QQ-A-225/9	SAE-AMS4133	SAE-AS5132
SAE-AMS-QQ-A-367	SAE-AMS4141	SAE-AS5187
SAE-AMS-QQ-P-416	SAE-AMS4339	SAE-AS8879

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2020-038)

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